

A Technical Introduction to Decision tree

Section 1

Decision tree using flowchart symbols Commonly a decision tree is drawn using flowchart symbols as it is easier for many to read and understand. Note there is a conceptual error in the "Proceed" calculation of the tree shown below; the error relates to the calculation of "costs" awarded in a legal action.

Section 2

Once we have calculated the key metrics we can make some initial conclusions on the performance of the decision tree model built. The accuracy that we calculated was 71.60%. The accuracy value is good to start but we would like to get our models as accurate as possible while maintaining the overall performance. The sensitivity value of 19.64% means that out of everyone who was actually positive for cancer tested positive. If we look at the specificity value of 99.06% we know that out of all the samples that were negative for cancer actually tested negative. When it comes to sensitivity and specificity it is important to have a balance between the two values, so if we can decrease our specificity to increase the sensitivity that would prove to be beneficial. These are just a few examples on how to use these values and the meanings behind them to evaluate the decision tree model and improve upon the next iteration.

Section 3

The decision tree illustrates that when sequentially distributing lifeguards, placing a first lifeguard on beach #1 would be optimal if there is only the budget for 1 lifeguard. But if there is a budget for two guards, then placing both on beach #2 would prevent more overall drownings.

Section 4

They are unstable, meaning that a small change in the data can lead to a large change in the structure of the optimal decision tree. They are often relatively inaccurate. Many other predictors perform better with similar data. This can be remedied by replacing a single decision tree with a random forest of decision trees, but a random forest is not as easy to interpret as a single decision tree. For data including categorical variables with different numbers of levels, information gain in decision trees is biased in favor of those attributes with more levels. Calculations can get very complex, particularly if many values are uncertain and/or if many outcomes are linked.

Section 5

We will set D, which is the depth of the decision tree we are building, to three ($D = 3$). We also have the following data

set of cancer and non-cancer samples and the mutation features that the samples either have or do not have. If a sample has a feature mutation then the sample is positive for that mutation, and it will be represented by one. If a sample does not have a feature mutation then the sample is negative for that mutation, and it will be represented by zero. To summarize, C stands for cancer and NC stands for non-cancer. The letter M stands for mutation, and if a sample has a particular mutation it will show up in the table as a one and otherwise zero.

Section 6

if condition1 and condition2 and condition3 then outcome. Decision rules can be generated by constructing association rules with the target variable on the right. They can also denote temporal or causal relations.

Section 7

Increasing the number of levels of the tree The accuracy of the decision tree can change based on the depth of the decision tree. In many cases, the tree's leaves are pure nodes. When a node is pure, it means that all the data in that node belongs to a single class. For example, if the classes in the data set are Cancer and Non-Cancer a leaf node would be considered pure when all the sample data in a leaf node is part of only one class, either cancer or non-cancer. A deeper tree is not always better when optimizing the decision tree. A deeper tree can influence the runtime in a negative way. If a certain classification algorithm is being used, then a deeper tree could mean the runtime of this classification algorithm is significantly slower. There is also the possibility that the actual algorithm building the decision tree will get significantly slower as the tree gets deeper. If the tree-building algorithm being used splits pure nodes, then a decrease in the overall accuracy of the tree classifier could be experienced. Occasionally, going deeper in the tree can cause an accuracy decrease in general, so it is very important to test modifying the depth of the decision tree and selecting the depth that produces the best results. To summarize, observe the points below, we will define the number D as the depth of the tree. Possible advantages of increasing the number D:

Section 8

Optimizing a decision tree A few things should be considered when improving the accuracy of the decision tree classifier. The following are some possible optimizations to consider when looking to make sure the decision tree model produced makes the correct decision or classification. Note that these things are not the only things to consider but only some.

Section 9

The tree using information gain has the same results when using the phi function when calculating the accuracy. When we classify the samples based on the model using information gain we get one true positive, one false positive, zero false negatives, and four true negatives. For the model using the phi function we get two true positives, zero false positives, one false negative, and three true negatives. The next step is to evaluate the effectiveness of the decision tree using some key metrics that will be discussed in the evaluating a decision tree section below. The metrics that will be discussed below can help determine the next steps to be taken when optimizing the decision tree.

Section 10

This is the phi function formula. The phi function is maximized when the chosen feature splits the samples in a way that produces homogenous splits and have around the same number of samples in each split.

Section 11

Runtime issues Decrease in accuracy in general Pure node splits while going deeper can cause issues. The ability to test the differences in classification results when changing D is imperative. We must be able to easily change and test the variables that could affect the accuracy and reliability of the decision tree-model.